SQL Master Librarian: Managing & Connecting Data

Comprehensive Lesson Guide & Worksheet | Planrs Academy

Topic: SQL (INSERT, UPDATE, DELETE, JOIN) **Target Audience:** Beginners / Middle School **Subject:** Computer Science & Databases

1. Introduction: Real-World Databases

Have you ever wondered how food delivery apps like **Zomato** know exactly what meal you ordered, your home address, and which delivery person is bringing your food? Or how a school administration system keeps track of thousands of students without mixing up their exam marks?

The secret lies in **Relational Databases**. Databases store information in neat, organized structures called **Tables**. A table is structured like a grid with rows and columns—just like a class attendance register!

Core Concepts recap:

- **Table:** A collection of related data organized in rows (records) and columns (attributes).
- **Primary Key:** A unique identifier for each row in a table (e.g., your Student Roll Number or Aadhaar/ID Number). No two records can share the same Primary Key!

2. Data Manipulation Commands (DML)

As a *Master Librarian* of data, you need powerful commands to maintain, modify, and clean your database tables. Here are the three main data modification commands:

1. Adding Data: `INSERT INTO`

When a new item or student arrives, we insert a new row into the table.

```
INSERT INTO Players (PlayerID,
Name, Runs)
VALUES (7, 'Virat', 0);
```

Effect: Adds a new 7th player named 'Virat' starting with 0 runs.

2. Modifying Data: `UPDATE`

When information changes (e.g., scores increase, address updates), we modify existing rows.

```
UPDATE Players
SET Runs = 75
WHERE PlayerID = 7;
```

Effect: Updates Virat's runs from 0 to 75.

3. Removing Data: `DELETE`

When data is no longer needed (e.g., a player retires, a book is lost), we remove the record.

```sql
DELETE FROM Players WHERE PlayerID = 3;
```

Effect: Completely removes the record with PlayerID = 3 from the table.

CRITICAL SAFETY RULE FOR UPDATE & DELETE:

Always specify a `WHERE` clause! If you write `DELETE FROM Players;` without a `WHERE` clause, **ALL records in the table will be permanently wiped out!** Similarly, omitting `WHERE` in `UPDATE` changes every row in the table.

3. Connecting Data Across Tables: `JOIN`

In big applications, data is split into multiple specialized tables to avoid repeating information. To answer complex questions, we use **JOIN** to connect related tables using a shared column (usually a Primary Key or Foreign Key).

Students Table

RollNo	Name	Class
101	Priya	6B
102	Rahul	6B
103	Ananya	6A

Library Table

RollNo	BookTitle	ReturnDate
101	**Panchatantra**	**20-Aug**
103	Harry Potter	22-Aug

Querying Joined Data: Who borrowed 'Panchatantra'?

```sql
SELECT Students.Name, Library.BookTitle
FROM Students
JOIN Library ON Students.RollNo = Library.RollNo
WHERE Library.BookTitle = 'Panchatantra';
```

Output Result: Priya — Panchatantra

Student Worksheet & Practice Questions

Test your SQL skills as the Master Librarian!

Instructions: Read each scenario carefully. Select or write the correct SQL command (`INSERT`, `UPDATE`, `DELETE`, or `JOIN`) or write the full SQL query where requested.

Section A: Conceptual & Scenario Assessment

Question 1

Scenario: A new student named Priya has newly joined Class 6B. You need to add her information to the school database. Which command should you use?

(A) UPDATE

(B) INSERT

(C) DELETE

(D) JOIN

Question 2

Scenario: A student named Priya has moved to a new home. You need to update her residential address in the school system without changing any other student records. Which command will you execute?

(A) JOIN

(B) DELETE

(C) UPDATE

(D) INSERT

Question 3

Scenario: A library book titled "Panchatantra" was damaged beyond repair and must be removed from the catalog database permanently. Which command perform this task?

(A) DELETE

(B) INSERT

(C) UPDATE

(D) JOIN

Question 4

Scenario: You have a "Students" table with student details and an "ExamScores" table with marks. You are asked to generate a report showing student names alongside their respective test marks. Which command allows you to link these two tables together?

(A) UPDATE

(B) JOIN

(C) INSERT

(D) DELETE

Question 5

Scenario: The sports coach wants to add a newly recruited player, 'Rohit', with PlayerID = 8 into the "Players" database table. Which command accomplishes this?

(A) INSERT

(B) UPDATE

(C) DELETE

(D) JOIN

Section B: Applied Query & Technical Logic

Question 6

Write a complete SQL query to insert a new book into the `Library` table with the following values: `BookID = 501`, `BookTitle = 'The Secret Garden'`, and `Author = 'Frances Hodgson Burnett'`.

Question 7

What happens if a librarian executes the query `DELETE FROM Library;` without adding a `WHERE` clause? Explain the outcome.

Question 8

Given two tables — `Orders (OrderID, CustomerID, ItemName)` and `Customers (CustomerID, CustomerName, Address)` — write an SQL `JOIN` query to display the `CustomerName` and `ItemName` for all orders.